Scalarization methods with a characterizing function of an ordering cone in set optimization

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Contents

Introduction

Preliminaries

Recap: Set-Valued Fan-Takahashi Inequalities with Set-Relations

Scalarization Function: Hiriart-Urruty Oriented Distance

Conclusion

Introduction

Introduction

In 2024, Dechboon and Tanaka examine whether composite functions of a set-valued map and Tammer's scalarization function, which is an extended real-valued function, are lower semicontinuous.

Last year, in this workshop, we talked about **Set-Valued Fan-Takahashi Inequalities with Set-Relations Based on Scalarization Methods.**

- Georgiev and Tanaka [3] extended Fan-Takahashi minimax inequality to the form of set-valued maps.
- Kuwano, Tanaka, and Yamada [4] constructed the result of four types of set-valued minimax inequalities with set relations.

In this talk, we explain scalarization functions from another perspective.

Preliminaries

Let X be a topological space, Y a real topological vector space, and θ_Y be a zero vector in Y. Define that $\mathcal{P}_0(Y)$ is the set of all nonempty subsets of Y. The sets of neighborhoods of $x \in X$ and $y \in Y$ is denoted by $\mathcal{N}_X(x)$ and $\mathcal{N}_Y(y)$, respectively.

For given $A, B \in \mathcal{P}_0(Y)$ and $t \in \mathbb{R}$, the algebraic sum $A + B$ and the scalar multiplication tA are denoted as follows:

$$A + B := \{a + b \mid a \in A, b \in B\}, \quad tA := \{ta \mid a \in A\}.$$

Definition

For $A, B \in \mathcal{P}_0(Y)$, we define two binary relations on $\mathcal{P}_0(Y)$:

$$A \preccurlyeq_1 B \overset{\text{def}}{\iff} A \cap B \neq \emptyset \quad \text{and} \quad A \preccurlyeq_2 B \overset{\text{def}}{\iff} B \subset A.$$

Preliminaries (Lower Semicontinuity)

Definition

Let $f : Y \to \mathbb{R} \cup \{\pm\infty\}$ and $y_0 \in Y$. We say that f is lower semicontinuous (l.s.c. shortly) at y_0 if

$$\forall r < f(y_0), \exists V \in \mathcal{N}_Y(y_0) \text{ s.t. } r < f(y), \forall y \in V;$$

Definition [1]

Let $F : X \to \mathcal{P}_0(Y)$, $x_0 \in X$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$ and $C \subset Y$ a convex cone. We say that F is $(\preccurlyeq, C)$-continuous at x_0 if

$$\forall W \subset Y, W \text{ open}, W \preccurlyeq F(x_0), \exists V \in \mathcal{N}_X(x_0) \text{ s.t. } W + C \preccurlyeq F(x), \forall x \in V.$$

Remark

As special cases, $(\preccurlyeq_1, C)$-continuity and $(\preccurlyeq_2, C)$-continuity coincide with " C-lower continuity " and " C-upper continuity " for set-valued maps, respectively.

Definition [1]
Let $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $A_0 \in \mathcal{P}_0(Y)$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and C a convex cone in Y with $C \neq Y$. Then, we say that φ is $(\preccurlyeq, C)$-lower semicontinuous at A_0 if

$$\forall r < \varphi(A_0), \exists W \in \mathcal{P}_0(Y), W \text{ open, s.t. } W \preccurlyeq A_0 \text{ and } r < \varphi(A), \forall A \in U(W + C, \preccurlyeq);$$

where $U(V, \preccurlyeq) \coloneqq \{A \in \mathcal{P}_0(Y) \mid V \preccurlyeq A\}$.

Theorem [1]
Let $F\colon X \to \mathcal{P}_0(Y)$, $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $x_0 \in X$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and C a convex cone. If F is $(\preccurlyeq, C)$-continuous at x_0 and φ is $(\preccurlyeq, C)$-lower semicontinuous at $F(x_0)$, then $(\varphi \circ F)$ is lower semicontinuous at x_0.

Definition

Let C be a convex cone in Y. A scalarization function φ is $(\preccurlyeq_C^{(j)})$-monotone if for any $A, B \in \mathcal{P}_0(Y)$ with $A \preccurlyeq_C^{(j)} B$, $\varphi(A) \leq \varphi(B)$.

Definition

Let C be a convex cone in Y. A scalarization function φ is $(\preccurlyeq_{\operatorname{int} C}^{(j)})$-monotone if for any $A, B \in \mathcal{P}_0(Y)$ with $A \preccurlyeq_{\operatorname{int} C}^{(j)} B$, $\varphi(A) < \varphi(B)$.

Recap: Set-Valued Fan-Takahashi Inequalities with Set-Relations

Theorem (Fan-Takahashi [5])
Let X be a nonempty compact convex subset of a Hausdorff topological vector space and $f : X \times X \to \mathbb{R}$. If f satisfies the following conditions:

1. for each fixed $y \in X$, $f(\cdot, y)$ is lower semicontinuous,
2. for each fixed $x \in X$, $f(x, \cdot)$ is quasi concave,
3. $f(x, x) \leq 0$ for all $x \in X$,

then there exists $\bar{x} \in X$ such that $f(\bar{x}, y) \leq 0$ for all $y \in X$.

Recap: Set-Valued Fan-Takahashi Inequalities with Set-Relations

Assumption $\Longrightarrow$ **Conclusion**

- for set-valued maps

- for scalarization functions

$$\varphi \in \Phi(\preccurlyeq_C^{(j)}, \preccurlyeq, C')$$

or

$$\varphi \in \Phi(\preccurlyeq_{\operatorname{int} C}^{(j)}, \preccurlyeq, C')$$

there exists $x \in X$ such that

$$F(\bar{x}, y) \preccurlyeq_C^{(j)} \{\theta_Y\}$$

or

$$\{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C} F(\bar{x}, y)$$

for all $y \in X$

Assumption $\Longrightarrow$ **Conclusion**

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$$F(\bar{x}, y) \preccurlyeq_C^{(j)} \{\theta_Y\}$$

or

$$\{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C} F(\bar{x}, y)$$

for all $y \in X$

Scalarization Functions (1/2)

Let $\varphi : \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $\preccurlyeq_C^{(j)} (j = 1, 2U, 2L, 3U, 3L, 4)$ set-relations, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and $C' \subset Y$ a convex cone. To generalize four types of set-valued minimax inequalities [4], we provide a new class of scalarization functions that satisfy;

(sc) φ is $(\preccurlyeq, C')$-lower semicontinuous;

(qv) φ is quasi concave.

Scalarization Functions (2/2)

In addition, we give necessary conditions between values of scalarization functions and set-relations as follows;

(i) φ is $(\preccurlyeq_C^{(j)})$-monotone,
(ii) $\varphi(A) \leq 0 \Rightarrow A \preccurlyeq_C^{(j)} \{\theta_Y\}$ for any $A \in \mathcal{P}_0(Y)$,
(i)'φ is $(\preccurlyeq_{\mathrm{int}\, C}^{(j)})$-monotone,
(ii)'$\varphi(A) > 0 \Rightarrow \{\theta_Y\} \preccurlyeq_{\mathrm{int}\, C}^{(j)} A$ for any $A \in \mathcal{P}_0(Y)$, and
(iii) $\varphi(\{\theta_Y\}) \leq 0$.

Supposed that φ satisfies conditions (sc) (qv), (i), (ii), and (iii), it is written as $\varphi \in \Phi(\preccurlyeq_C^{(j)}, \preccurlyeq, C')$. On the other hand, if φ satisfies conditions (sc) (qv), (i)', (ii)', and (iii), we write the notation as $\varphi \in \Phi(\preccurlyeq_{\mathrm{int}\, C}^{(j)}, \preccurlyeq, C')$.

Set-Valued Fan-Takahashi Inequalities with Set-Relations (1/2)

Theorem 1

Let X be a nonempty compact convex subset of a Hausdorff topological vector space, Y a real topological vector space, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, C a convex cone in Y, C' a convex cone in Y, $\preccurlyeq_C^{(j)}$ $(j = 1, 2U, 2L, 3U, 3L, 4)$ set-relations, $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, and $F\colon X \times X \to \mathcal{P}_0(Y)$. For the scaralization function $\varphi \in \Phi(\preccurlyeq_C^{(j)}, \preccurlyeq, C')$, if F satisfies the following conditions:

1. $(\varphi \circ F)(x, y) \in \mathbb{R}$ for all $x, y \in X$,
2. for each fixed $y \in X$, $F(\cdot, y)$ is $(\preccurlyeq, C')$-continuous,
3. for each fixed $x \in X$, $F(x, \cdot)$ is type (j) naturally quasi C-concave,
4. for all $x \in X$, $F(x, x) \preccurlyeq_C^{(j)} \{\theta_Y\}$,

then there exists $\bar{x} \in X$ such that $F(\bar{x}, y) \preccurlyeq_C^{(j)} \{\theta_Y\}$ for all $y \in X$.

Set-Valued Fan-Takahashi Inequalities with Set-Relations (2/2)

Theorem 2

Let X be a nonempty compact convex subset of a Hausdorff topological vector space, Y a real topological vector space, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, C a convex cone in Y, C' a convex cone in Y, $\preccurlyeq_C^{(j)}$ $(j = 1, 2U, 2L, 3U, 3L, 4)$ set-relations, $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, and $F\colon X \times X \to \mathcal{P}_0(Y)$. For the scaralization function $\varphi \in \Phi(\preccurlyeq_{\mathrm{int}\, C}^{(j)}, \preccurlyeq, C')$ if F satisfies the following conditions:

1. $(\varphi \circ F)(x, y) \in \mathbb{R}$ for all $x, y \in X$,
2. for each fixed $y \in X$, $F(\cdot, y)$ is $(\preccurlyeq, C')$-continuous,
3. for each fixed $x \in X$, $F(x, \cdot)$ is type (j) naturally quasi C-concave,
4. for all $x \in X$, $\{\theta_Y\} \preccurlyeq_{\mathrm{int}\, C}^{(j)} F(x, x)$,

then there exists $\bar{x} \in X$ such that $\{\theta_Y\} \preccurlyeq_{\mathrm{int}\, C}^{(j)} F(\bar{x}, y)$ for all $y \in X$.

Scalarization Function: Hiriart-Urruty Oriented Distance

Definition (Hiriart-Urruty Oriented distance) [6]
Let Y be a real normed vector space. For a set $A \subset Y$, let the oriented distance function $\Delta_A \colon Y \to \mathbb{R} \cup \{\pm\infty\}$ be defined by

$$\Delta_A(y) \coloneqq d_A(y) - d_{Y \setminus A}(y),$$

$d_A(y) = \inf\{\|y - z\| \mid z \in A\}$, $d_\emptyset(y) \coloneqq +\infty$, and $\|y\|$ denotes the norm of y in Y.

Definition (Generalized Hiriart-Urruty Oriented distance) [6]
For the set $A \in Y$, let the functions $\mathcal{D}_A^+ \colon \mathcal{P}(Y) \to \mathbb{R} \cup \{\pm\infty\}$ and $\mathcal{D}_A^- \colon \mathcal{P}(Y) \to \mathbb{R} \cup \{\pm\infty\}$ be defined as

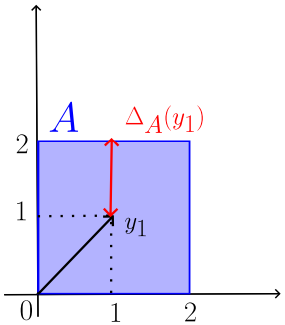
$$\mathcal{D}_A^{(1)}(B) \coloneqq \sup\{\Delta_A(b) \mid b \in B\},\ B \in \mathcal{P}(Y),$$
$$\mathcal{D}_A^{(2)}(B) \coloneqq \inf\{-\Delta_A(b) \mid b \in B\} = -\mathcal{D}_A^{(1)}(B),\ B \in \mathcal{P}(Y).$$

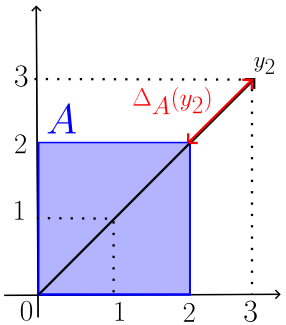
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Example

Let $Y = \mathbb{R}^2$, $A = [0,2] \times [0.2]$, $y_1 = (1,1)$, and $y_2 = (3,3)$. Then,

$$\Delta_A(y_1) = d_A(y_1) - d_{Y \setminus A}(y_1) = 0 - 1 = -1,$$
$$\Delta_A(y_2) = d_A(y_2) - d_{Y \setminus A}(y_2) = \sqrt{2} - 0 = \sqrt{2}.$$





Lemma

Let C be a proper closed convex cone with $\operatorname{int} C \neq \emptyset$ in $\hat{Y}$. Then, for any set in $\{B \in \mathcal{P}_0(\hat{Y}) \mid B \text{ is } C\text{-compact}\}$, $\mathcal{D}_C^{(2)}(B) \in \Phi(\preccurlyeq^{(3L)}_{\operatorname{int} C}, \preccurlyeq_2, C)$.

Proposition

Let X be a nonempty compact convex subset of a topological vector space, $\hat{Y}$ a real normed vector space, C a proper closed convex cone in $\hat{Y}$ with $\operatorname{int} C \neq \emptyset$, and, $F\colon X \times X \to \mathcal{P}_0(\hat{Y})$. If F satisfies the following conditions:

1. F is C-compact on $X \times X$,
2. for each fixed $y \in X$, $F(\cdot, y)$ is $(\preccurlyeq_2, C)$-continuous (that is, C-upper continuous),
3. for each fixed $x \in X$, $F(x, \cdot)$ is type (j) naturally quasi C-concave,
4. for all $x \in X$, $\{\theta_{\hat{Y}}\} \preccurlyeq^{(3L)}_{\operatorname{int} C} F(x, x)$,

then there exists $\bar{x} \in X$ such that $\{\theta_{\hat{Y}}\} \preccurlyeq^{(3L)}_{\operatorname{int} C} F(\bar{x}, y)$ for all $y \in X$.

Scalarization Functions (again)

In addition, we give necessary conditions between values of scalarization functions and set-relations as follows;

(i) φ is $(\preccurlyeq_C^{(j)})$-monotone,
(ii) $\varphi(A) \leq 0 \Rightarrow A \preccurlyeq_C^{(j)} \{\theta_Y\}$ for any $A \in \mathcal{P}_0(Y)$,
(i)'φ is $(\preccurlyeq_{\mathrm{int}\, C}^{(j)})$-monotone,
(ii)'$\varphi(A) > 0 \Rightarrow \{\theta_Y\} \preccurlyeq_{\mathrm{int}\, C}^{(j)} A$ for any $A \in \mathcal{P}_0(Y)$, and
(iii) $\varphi(\{\theta_Y\}) \leq 0$.

Supposed that φ satisfies conditions (sc) (qv), (i), (ii), and (iii), it is written as $\varphi \in \Phi(\preccurlyeq_C^{(j)}, \preccurlyeq, C')$. On the other hand, if φ satisfies conditions (sc) (qv), (i)', (ii)', and (iii), we write the notation as $\varphi \in \Phi(\preccurlyeq_{\mathrm{int}\, C}^{(j)}, \preccurlyeq, C')$.

Scalarization Functions (again)

In addition, we give necessary conditions between values of scalarization function and set-relations as follows;

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Characterizing functions

Definition [2]

Let C be a convex cone in Y and $f : Y \to \mathbb{R} \cup \{\pm\infty\}$. We say that f is a characterizing function of the ordering cone $-C$ if

$$f(y) \leq 0 \Leftrightarrow y \in -C \quad \text{for all } y \in Y.$$

Definition

Let C be a convex cone in Y and $\preccurlyeq_C^{(j)}$ $(j = 1, 2U, 2L, 3U, 3L, 4)$ set-relations, For a scalarization function $\varphi : \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, we consider a class of scalarization functions satisfying following three conditions is denoted as $\Phi_0(\preccurlyeq_C^{(j)})$;

1. φ is $(\preccurlyeq_C^{(j)})$-monotone,
2. $\varphi(A) \leq 0 \Rightarrow A \preccurlyeq_C^{(j)} \{\theta_Y\}$ for any $A \in \mathcal{P}_0(Y)$,
3. $\varphi(\{\theta_Y\}) \leq 0$.

Lemma

Let C be a convex cone in Y, $\preccurlyeq_C^{(j)}$ $(j = 1, 2U, 2L, 3U, 3L, 4)$ set-relations, Consider a scalarization function $\varphi \in \Phi_0(\preccurlyeq_C^{(j)})$. Then, if we define $f(y) = \varphi(\{y\})$ for any $y \in Y$, f is a characterizing function of $-C$.

Lemma

Let C be a closed convex cone in Y and $f : Y \to \mathbb{R} \cup \{\pm\infty\}$ a characterizing function of $-C$. Define $\varphi_s : \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$ by

$$\varphi_s(A) = \sup_{a \in A} f(a) \quad \text{for all } A \in \mathcal{P}_0(Y).$$

If f satisfies that for all $a, b \in Y$ with $a \leq_C b$, $f(a) \leq f(b)$, then φ_s is an element of $\Phi_0(\preccurlyeq_C^{(3U)})$.

Conclusion

Conclusion

- We can give an example of interior ordering type of set-valued minimax inequalities. However, for ordinary ordering type of set-valued minimax inequalities, these examples are not provided here. As the result between scalarization functions and characterizing functions can be a key element for establishing a new example.
- The form of composite functions appear in some fields. Thus, studying continuity of set-valued maps, we find other applications except for Fan-Takahashi minimax inequality.

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